

Gene Pulser® Electroporation

Cell Type	Bacterial, gram positive	Molecules Electroporated	DNA: pBR325, 9 to 10kB; original plasmid from <i>Cornibacterium</i> , to 10 kB	5
Species Used	<i>Corynebacterium</i> , unspecified strain ; <i>Brevibacterium flavum</i> , ATCC 21475			

Before the Pulse

Cell Growth Medium	L. Broth with 1% glucose	Growth Phase at Harvest	O.D. (600) =0.5 to 1.0
		Pre-pulse Incubation	ice

Wash Solution Cold water

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature	25 °C		
Electroporation Medium	10% glycerol solution	Cuvette Gap	0.4 cm
Cell Density	1 to 3 x 10 (10) cells / ml	Voltage	2.5 kV
Volume of Cells	40 µl	Field Strength	6.25 kV/cm
DNA Concentration	50 to 100 µg / ml		
DNA Resuspension Buffer	TE buffer (10 mM Tris, 1 mM EDTA, pH 8.0)	Capacitor	125 µF
Volume of DNA	2 µl	Resistor	(Pulse Controller) none
After the Pulse		Time Constant	4 to 5 msec
Outgrowth Medium	SOC		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

SOC: 2% Bacto tryptone, 0.5% Bacto yeast extract, 10mM NaCl, 2.5mM KCl, 10 mM MgCl₂, 10 mM MgSO₄, 20 mM glucose.
LB: 1% Bacto tryptone, 0.5% Bacto yeast extract, 0.5% NaCl.

Outgrowth Temperature	30 °C
Length of Incubation	1 hr. to 2 hr.
Selection Method or Assay Used	Various antibiotics
Electroporation Efficiency	10 (6) to 10 (7) transformants / µg DNA
Per Cent Survival	Not given

Name of Submitter Not given

Institution Address Asahi Chemical Industry Co-ltd.
 Foods R&D Section
 Asahi omachi 6-2700
 Noheoka, Miyazaki
 Japan

Survey Number
 058